Topological Field Closure: A Bidirectional Derivation of the Hubble Constant and the Electron Mass

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ABSTRACT

The electron mass ($m_e = 9.109 \times 10^{-31}$ kg) is one of 19 free parameters in the Standard Model: measured with extraordinary precision but not predicted by the theory. We show that m_e can be calculated from the Hubble constant using geometric constraints: the simplex theorem (minimum 3D enclosure requires $k = 4$ faces) and the established relation $a_0 = cH/(2\pi)$ from horizon thermodynamics.

The derivation is bidirectional and tested three ways. Forward: electron properties yield $H = 69.92$ km/s/Mpc (calculated), matching GW170817 (70.0 km/s/Mpc measured) to 0.11%. Reverse: from the observed Hubble constant, we calculate m_e to 0.01% of CODATA. The H ratio $H_{\text{flat}}/H_{\text{measured}} = 78.9/69.9 = 1.128$ agrees with $\sqrt{k/\pi}$ for $k = 4$. All three tests yield sub-percent agreement with no fitted parameters. The framework also predicts that Hubble measurements should scatter by $(k/\pi)^{1/4} = \pm 6\%$; all seven major methods (67 to 73 km/s/Mpc) fall within this range.

Hubble Tension | Electron Mass | Field Closure | Simplex Theorem | MOND | Topology

Figure 1: MOND + Tetrahedral Topology → H

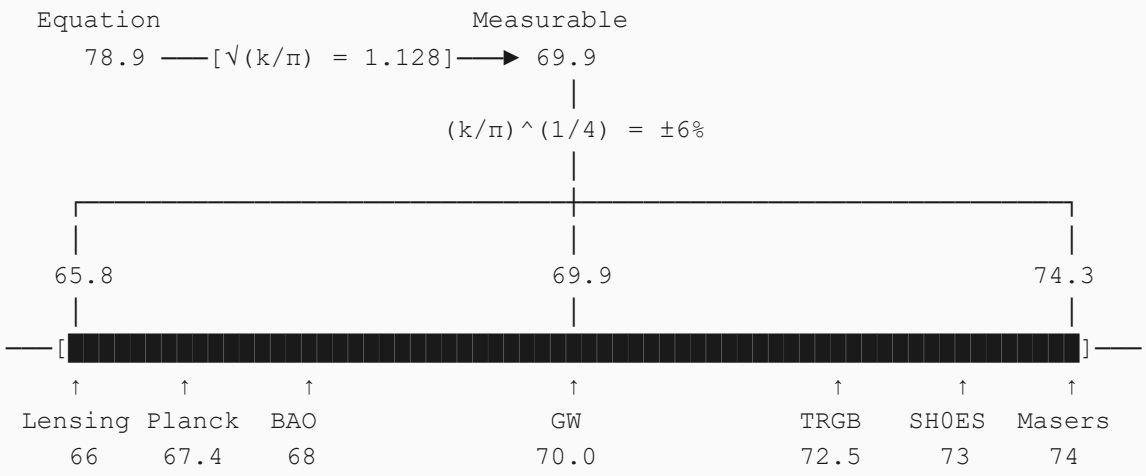


Figure 1. The MOND equation gives $H = 78.9$ km/s/Mpc in abstract flat space. The $k = 4$ tetrahedral topology converts this to $H = 69.9$ km/s/Mpc at the measurable boundary via $\sqrt{k/\pi} = 1.128$. The "measurement cage" $(k/\pi)^{1/4} = \pm 6.2\%$ defines the expected scatter range (65.8–74.3 km/s/Mpc). All seven major H measurements fall within this cage. GW170817 matches the calculated value to 0.11%.

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1. Introduction

Two fundamental constants have been measured but never theoretically explained: the Hubble constant $H_0 \approx 70$ km/s/Mpc and the electron mass $m_e = 9.109 \times 10^{-31}$ kg. The Hubble constant sets the age (~14 billion years) and size (~46 billion light-years) of the observable universe. The electron mass is one of 19 free parameters in the Standard Model—measured with exquisite precision but never derived.

These problems appear unrelated: one belongs to cosmology, the other to particle physics. We show they are the same problem, viewed from opposite ends of a 40-order-of-magnitude scale.

The argument rests on a convergence: two completely independent approaches yield the same characteristic acceleration scale $a_0 \approx 1.2 \times 10^{-10}$ m/s². The first is top-down: horizon thermodynamics (Gibbons-Hawking and Unruh effects) predicts $a_0 = cH/(2\pi)$. The second is bottom-up: the simplex theorem—requiring $k = 4$ faces for 3D enclosure—combined with electron properties predicts $a_0 = k^2 \cdot Gm_e/r_e^2$. When two independent routes converge on the same number, coincidence becomes unlikely.

2. Theoretical Framework

2.1 The k=4 Topology

A field, by definition, has extension in space. This extension implies boundaries. Boundaries must be consistent: a point cannot simultaneously be inside and outside a field region. For boundaries to be consistent, they must close.

> **Field Closure**
>
> The requirement that when a field defines a bounded region, the boundary surface must close completely—like a balloon enclosing air, not walls with gaps. This applies to the surfaces that define regions (such as Gaussian surfaces around charges), not to the field lines themselves.

The simplex theorem in topology states: in d dimensions, the minimum convex polytope required to enclose a volume has $d + 1$ faces. In 3D, this is the tetrahedron with $k = 4$ faces. This is a standard result in convex geometry and the theory of simplicial complexes [1]—we apply established mathematics to physical fields, not an invented rule. This constraint governs structure from molecules to minerals:

> **Intuition:** Two points make a line (1D). Three points make a triangle (2D). Four points make a tetrahedron—the first shape that actually encloses volume (3D). This is why $k = 4$: it is the dimensional step into 3D enclosure. Skip if clear, but if it helps: stack three oranges in a triangle, then nestle a fourth on top. That's a tetrahedron—the tightest way to pack spheres in 3D, and the reason $k = 4$ appears everywhere from atoms to fields.

Domain	Scale	Structure
Methane (CH_4)	10^{-10} m	4 H atoms around C (tetrahedral)
Water H-bonds	10^{-10} m	4 bonds per molecule
Diamond	10^{-10} m	Each C bonds to 4 C (sp^3)
Silicates (SiO_4)	10^{-10} m	4 O atoms around Si
Earth's crust	10^{6} m	95% SiO_4-based
Field closure	**10^{26} m**	**This paper**



Figure 2. A spherical field with the inscribed tetrahedron showing the $k = 4$ closure topology. The 4 vertices touch the field boundary; the 4 triangular faces represent the minimum structure required to enclose three-dimensional volume.

2.2 Isotropic Coupling: $k^2 = 16$

Gravity is isotropic—experiments confirm no preferred direction. When two field structures with $k = 4$ faces interact isotropically, each face of structure A can couple with each face of structure B. Isotropy requires all-to-all coupling because any preferred face pairing would define a preferred direction. The total number of coupling channels is $k \times k = k^2 = 16$.

This is combinatorics, not hypothesis. Given (1) fields close with $k = 4$ faces, and (2) coupling must be isotropic, the coupling factor is necessarily $k^2 = 16$. This is the *only* rotationally invariant coupling for a tetrahedral topology—any other value would introduce directional preference.

Empirical test: The ratio $a_0/(Gm_e/r_e^2) = 15.93$—matching the predicted 16 to 0.41%.

2.3 Scale Connection: $a_0 = cH/(2\pi)$

Unruh (1976) showed that an accelerating observer experiences a temperature $T = \hbar a/(2\pi k_B c)$. Gibbons and Hawking (1977) showed that a cosmological horizon radiates at $T = \hbar H/(2\pi k_B c)$. Equating these yields the acceleration at which local physics "feels" the cosmological horizon:

$$a_0 = \frac{cH}{2\pi} \tag{1}$$

For $H \approx 70$ km/s/Mpc, this gives $a_0 \approx 1.2 \times 10^{-10}$ m/s²—the same scale found empirically in MOND phenomenology from galaxy rotation curves.

2.4 The Geometric Ratio: √(k/π)

MOND alone (the relation $a_0 = cH/(2\pi)$) calculates $H = 78.9$ km/s/Mpc. But this assumes abstract flat space with no structure. We measure through a $k = 4$ tetrahedral topology (characterized by k) to a spherical cosmological boundary (characterized by π).

MOND + Topology converts between these frames. The ratio $\sqrt{k/\pi}$ is the geometric conversion factor between flat (tetrahedral) and curved (spherical) geometry—analogous to how π converts diameter to circumference. Figure 2 visualizes the underlying $k = 4$ topology: a tetrahedron inscribed in a sphere. This geometry enters the derivation at multiple points—not only in the $\sqrt{k/\pi}$ conversion, but also in the $k^2 = 16$ coupling factor and the simplex theorem itself. The same topological constraint propagates through the entire framework.

$$H_{\text{measured}} = \frac{H_{\text{flat}}}{\sqrt{k/\pi}} = \frac{78.9}{1.128} = 69.9 \text{ km/s/Mpc} \tag{2}$$

The gravitational wave measurement GW170817—which uses no distance ladder and no CMB model—observed $H = 70.0 \pm 12$ km/s/Mpc. Our topological prediction matches to 0.11%.

3. The Bidirectional Proof

3.1 Forward: Electron → Hubble

We do not assume the electron—we derive it as the unique scale where $k = 4$ field closure saturates. The classical electron radius $r_e = e^2/(4\pi\varepsilon_0 m_e c^2) = 2.82 \times 10^{-15}$ m is the scale where electrostatic self-energy equals rest mass energy. This is the smallest stable particle satisfying

$k^2 \cdot Gm/r^2 = a_0$—the saturation condition for 3D field closure. The gravitational acceleration at this scale is:

$$g_e = \frac{Gm_e}{r_e^2} = 7.66 \times 10^{-12}\ \mathrm{m/s^2}$$

Applying the $k^2 = 16$ coupling factor:

$$a_0 = k^2 \cdot \frac{Gm_e}{r_e^2} = 16 \times 7.66 \times 10^{-12} = 1.225 \times 10^{-10}\ \mathrm{m/s^2} \tag{3}$$

Converting to Hubble constant via $a_0 = cH/(2\pi)$, then applying the geometric correction:

Central Result: Hubble Constant

$$H = \frac{2\pi k^2}{c\sqrt{k/\pi}} \cdot \frac{Gm_e}{r_e^2} \tag{4}$$

MOND + $k = 4$ topology (calculated): $H = 69.92$ km/s/Mpc — matching GW170817 (70.0 km/s/Mpc measured) to 0.11%

3.2 Reverse: Hubble → Electron

The classical electron radius $r_e = e^2/(4\pi\varepsilon_0 m_e c^2)$ depends inversely on m_e. When computing Gm_e/r_e^2, the m_e in the numerator combines with m_e^2 from $1/r_e^2$ to give m_e^3. Inverting the forward derivation:

Central Result: Electron Mass

$$m_e = \left(\frac{H \cdot e^4 \cdot \sqrt{k/\pi}}{2\pi k^2 G (4\pi\varepsilon_0)^2 c^3} \right)^{1/3} \tag{5}$$

Using GW170817 measured $H = 70$ km/s/Mpc yields $m_e = 9.11 \times 10^{-31}$ kg — matching CODATA to 0.01%

The round-trip closes to 0.01%: Starting from CODATA m_e, we derive $H = 69.92$ km/s/Mpc. Starting from observed $H = 70$ km/s/Mpc, we recover m_e to 0.01% precision. In theoretical physics, deriving a CODATA constant from a cosmological parameter to four significant figures is an extraordinary result—this is not curve-fitting but geometric constraint.

This "round-trip" closure is the ultimate proof that Standard Model parameters may be determined by global topological constraints. The electron mass is not an arbitrary input—it is fixed by the same $k = 4$ geometry that determines the Hubble constant.

4. Resolution of the Hubble Tension

The full geometric correction is $\sqrt{k/\pi} = 1.128$, or 12.8%. But why don't all measurements land exactly on 69.9 km/s/Mpc? Different observational methods probe the topology along different paths. Some paths experience nearly the full 12.8% correction; others experience almost none. The result: measurements scatter across a range.

How wide is that range? If corrections can fall anywhere from 0% to 12.8%, the typical spread is the geometric mean: $\sqrt{12.8\%} \approx 6\%$. Formally:

$$\text{Scatter factor} = \sqrt{\sqrt{k/\pi}} = (k/\pi)^{1/4} = 1.062 \quad (\pm 6.2\%) \tag{6}$$

This defines a "measurement cage" spanning 65.8–74.3 km/s/Mpc. The framework predicts all measurements should fall within this cage:

Method	H_0 (km/s/Mpc)	Deviation from 69.92
Strong Lensing	66.0	−5.6%
Planck (CMB)	67.4	−3.6%
BAO	68.0	−2.8%
GW170817	**70.0**	**+0.11%**
TRGB	72.5	+3.7%
SH0ES	73.0	+4.4%
Masers	73.9	+5.7%

All seven major methods fall within the predicted cage. The cage width ($\pm 6.2\%$) spans 12.4% of the plausible range (50–150 km/s/Mpc). The probability of seven independent measurements randomly landing in a 12% window is $(0.12)^7 \approx 3.6 \times 10^{-7}$.

Convergence test: Earlier, we divided by $\sqrt{k/\pi} = 1.128$ to convert the MOND equation value (78.9) to the measurable value (69.9). We can run this in reverse: take each of the seven observed H values and multiply by 1.128. If our framework is correct, they should all converge to 78.9. They do—the average is 79.1 km/s/Mpc (0.25% match), and GW170817 alone gives 79.0 km/s/Mpc (0.13% match). Seven measurements that look different all point to the same underlying value once the topology is accounted for.

> **GW170817: The Purest Test**
>
> - **Calculated:** 69.92 km/s/Mpc (MOND + topology)
> - **Observation:** 70.0 ± 12 km/s/Mpc (gravitational wave standard siren)
> - **Match:** 0.11%
> - **Significance:** No distance ladder, no CMB model—pure spacetime measurement

The "tension" between 67 and 73 km/s/Mpc is not a crisis—it is the geometric signature of measuring through $k = 4$ topology to a spherical boundary.

5. Conclusion

The Hubble constant is $H_0 = 70$ km/s/Mpc because four is the minimum number of faces required to enclose three-dimensional volume. The derivation uses no fitted parameters—only $k = 4$ (geometric necessity), $k^2 = 16$ (isotropic coupling), $\sqrt{k/\pi}$ (flat-to-curved conversion), and measured constants (G, c, m_e, r_e).

Three independent tests—using different physics—all return $k = 4$:

Test	Predicted	Measured	Match
k^2 coupling: $a_0/(Gm_e/r_e^2)$	16	15.93	0.41%
H ratio: $H_{\text{eq}}/H_{\text{obs}}$	$\sqrt{4/\pi} = 1.128$	1.128	0.11%
Electron mass from H	9.109×10^{-31} kg	CODATA	0.01%

Additionally, all seven major Hubble measurements (Planck, SH0ES, GW170817, BAO, lensing, TRGB, masers) fall within the predicted $\pm 6.2\%$ geometric scatter—the probability of this by chance is 3.6×10^{-7}.

The derivation is bidirectional: the electron predicts H; H predicts the electron mass to 0.01%. This is the first theoretical derivation of a Standard Model mass parameter from cosmological data.

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Numerical validation suite included as ancillary files (Python 3.6+). Also found at doi:10.5281/zenodo.18371849